USN | | | | | | | | | | [23CSBSMA22]

SECOND SEMESTER B.Tech EXAMINATION, JUNE - 2024

LINEAR ALGEBRA

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A	**05x02=10**		
1.	**a.**	Find the values of a, b, c and d which satisfy the matrix equation $\begin{bmatrix} a-2 & 2b+a \\ c-1 & 4d-6 \end{bmatrix} = \begin{bmatrix} 0 & -7 \\ 3 & 2d \end{bmatrix}$.	**L2**	**CO1**	**02**
	b.	Find the rank of a matrix $A = \begin{bmatrix} 1 & 3 & 4 & 5 \\ 2 & -1 & 1 & -3 \\ 3 & 2 & 5 & 2 \end{bmatrix}$.	**L2**	**CO2**	**02**
	c.	Let V be the set of all ordered pairs (a,b) where a,b are real numbers. Let x=(a_1,b_1) and y=(a_2,b_2) be two elements in V. Define the addition x+y = (a_1,b_1)+(a_2,b_2) = ($2a_1$-$3a_2$, b_1-b_2). Show that V is not a vector space.	**L2**	**CO4**	**02**
	d.	Calculate the eigenvalues of the matrix $A = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$	**L3**	**CO5**	**02**
	e.	Write the matrix representation of principal component analysis.	**L1**	**CO6**	**02**
		PART - B	**06x05=30**		
2.	**a.**	Find the inverse of the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$, if it exists.	**L3**	**CO1**	**05**
		OR			
	b.	Use determinant concept to find the area of the triangle whose vertices are (1,0), (2,2) and (4,3).	**L3**	**CO1**	**05**
3.	**a.**	Solve the system of equations $5x + y + 3z = 20$, $2x + 5y + 2z = 18$ and $3x + 2y + z = 14$ by Gauss-elimination method.	**L3**	**CO2**	**05**
		OR			
	b.	Solve the system of homogeneous equations: $2x - y + 2z = 0$, $4x + 3y + 3z = 0$, $12x + 7y + 10z = 0$.	**L3**	**CO2**	**05**
4.	**a.**	Given $x_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ and $x_2 = \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix}$ is a basis for a subspace of W. Use the Gram- Schmidt process to find an orthogonal basis for W.	**L3**	**CO4**	**05**
		OR			
	b.	Express the vector (4, 3, 10) as a linear combination of the vectors (1,2,-1),(2,3,4),(1,5,-3) in $V_3(R)$.	**L3**	**CO4**	**05**
5.	**a.**	Show that the set B={(1,1,0),(1,0,1),(0,1,1)} is a basis of vector space in $V_3(R)$.	**L3**	**CO4**	**05**
		OR			

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	b.	Find QR factorization of the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.	L3	CO4	05
6.	a.	For what value/s of b is the following matrix $A = \begin{bmatrix} 2 & -1 & b \\ -1 & 2 & -1 \\ b & -1 & 2 \end{bmatrix}$ positive semidefinite?	L2	CO5	05
		OR			
	b.	Is $A = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$ a unitary matrix? Use the condition AH = A^{-1} to verify it.	L2	CO5	05
7.	a.	Write an algorithm to do feature extraction of given data set using principal component analysis.	L2	CO6	05
		OR			
	b.	Write an algorithm to decompose a matrix A into singular value decomposition.	L2	CO6	05
		PART - C		06x10=60	
8.	a.	Use Cramer's rule to solve the system of equations $x - 2y - 2z = 3, 2x - 4y + 4z = 1$and $3x - 3y - 3z = 4$.	L3	CO1	10
		OR			
	b.	The upward speed v(t) of a rocket at time t is approximated by $v(t) = at^2 + bt + c$, $0 \le t \le 100$ where a, b, and c are constants. It has been found that the speed at times t = 3, t = 6, and t = 9 seconds are respectively, 64, 133, and 208 miles per second respectively. solve the system of equations by matrix method to find the speed at time t = 15 seconds.	L3	CO1	10
9.	a.	Solve the system of equations $3x + y + z = 4$, $x + 2y + 2z = 3$, and $2x + y + 3z = 4$ by LU decomposition method.	L3	CO3	10
		OR			
	b.	Investigate the values of λ and μ so that the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$ will have **i.** no solution, **ii.** a unique solution **iii.** an infinite number of solutions	L3	CO2	10
10.	a.	Find the four fundamental subspaces of the matrix $A = \begin{bmatrix} 1 & -2 & -1 & 3 & 2 \\ 2 & -2 & -3 & 6 & 1 \\ -1 & -4 & 4 & -3 & 7 \end{bmatrix}$	L3	CO4	10
		OR			
	b.	Let V be the set of all positive real numbers where addition is defined by $x + y = xy$ and scalar multiplication is defined by $kx = x^k$. Determine whether V is a vector space.	L3	CO4	10

11.	**a.**	If $T: R^2 \to R^2$ is a linear transformation such that $T(1, 0) = (1, 1)$ $and\, T(0, 1) = (-1, 2)$. Show that T maps the square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$ into a parallelogram.	**L3**	**CO5**	**10**
		OR			
	b.	Reduce the matrix $A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$ into Eigenvalue decomposition form.	**L3**	**CO5**	**10**
12.	**a.**	Find the singular value decomposition of the matrix $A = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix}$.	**L3**	**CO6**	**10**
		OR			
	b.	Compute the principal component of following data. CLASS 1 X = 2 , 3 , 4 ; Y = 1 , 3 , 4 CLASS 2 X = 5 , 6 , 7; Y = 6 , 7 , 8 using PCA algorithm	**L3**	**CO6**	**10**
13.	**a.**	Find the singular value decomposition of the matrix $A = \begin{pmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{pmatrix}$	**L3**	**CO6**	**10**
		OR			
	b.	Compute the principal component using PCA Algorithm for the two-dimensional patterns (2, 1), (3, 5), (4, 3), (5, 6), (6, 7), (7, 8).	**L3**	**CO6**	**10**